

Departmental systems unaffected

Sentinel: Model 2 deliverable (Unified Viewing & Metadata Analytics). Gujarat Police Innovation Challenge 2026, 25 September 2026. Sources: `deliverables/HLD.md` §2.6 (the guarantee to departments), §2.7 (two systems in one viewer), §3.3 (relayed, not recorded); `docs/architecture.md` Part A invariants, B2 and Part D.

The guarantee. Sentinel is consume-only. It pulls each camera's stream **read-only and once**, and fans it out inside the platform. It never publishes to a gateway, never calls a control API, never downloads stored footage and never writes to a departmental system. A department's cameras, VMS, storage and retention policy are exactly as they were before onboarding. The demonstrated system holds to this against the organisers' sandbox, and the source can be checked.

On the department's side	Sentinel	Enforced in
Read the live stream	Yes, read-only: RTSP over TCP, HLS where RTSP is blocked, never UDP	<code>ml/ingest/rtsp.py</code>
A second connection to the same camera	No. The analytics decoder stream-copies a local 20-second window as it reads; the Live Wall plays that window	<code>backend/app/routes_hls.py</code>
Publish or write to a camera, gateway or VMS	No. Sentinel publishes only to its own mediamtx on <code>127.0.0.1</code> (the stock sample feeds)	<code>scripts/replay_publish.py</code>
Call a control API (PTZ, recording, configuration)	No	root rule 6
Download or bulk-copy stored footage	No. The harvest of the organisers' recordings was cut	HLD Appendix B
Store video	Only on Sentinel's own disk: a self-overwriting 20-second relay window per analysed camera	HLD §1.4
Hold the department's credentials	Environment variables only; never in the database, a URL, a response or an unmasked log line	root rule 1

How each camera is viewed: two different systems in one viewer.

Camera	The Live Wall plays	Connection to the department's side
Sandbox camera analysed by the node (five, e.g. cam06)	the node's own pull, from its 20-second window	the single analytics pull, nothing more
Sandbox camera not analysed (the other 25)	the organisers' own CDN recording , relayed segment by segment at the current position on the shared timeline; the tile reads "CDN RECORDING"	none to the live gateway; one CDN request per segment however many people watch (in-memory cache, nothing on disk)
Local sample feed (<code>local101</code> – <code>local128</code> , stock footage)	Sentinel's own mediamtx HLS on <code>127.0.0.1</code>	none: Sentinel's own server

Why the other 25 show the CDN recording. The five analysed cameras already hold five RTSP sessions on the gateway, which appears to allow about six per account (suspected from a run in which two of eight cameras pulled no frames; not measured). A live-wall RTSP pull would be a second connection and would starve the analytics. The organisers' own HLS copy, relayed once, keeps one pull per camera and leaves the gateway alone.

What the relay fetches. Only from the configured CDN origin, and only segment names listed in the playlist it has just fetched. It sits behind the login and is rate-limited. While the CDN refuses requests, a circuit breaker with jittered backoff answers the tile at once instead of retrying into a ban. The only upstream request that is not a `GET` is the CDN's viewer login, which obtains the session cookie a browser would hold.

Health checks. An analysed camera is never probed; its health is the freshness of its relay window. Other sandbox cameras get a short, paced RTSP `ffprobe` over TCP, two at a time, every five minutes. The CDN is never used as a probe.

What a department provides. A reachable stream endpoint (RTSP or ONVIF, or an HLS gateway) and a read-only account, registered as a URL template without the credential. Nothing is installed or changed on its VMS. A system

reachable only through a vendor SDK, or with no inbound route, would be served by a department-side collector that dials out (Model 3, HLD §2.3; described, not built).

Stated limits. Sandbox departments and coordinates are seeded demonstration assignments. The 28 local feeds are stock footage at seeded coordinates (`sample-camera-dataset.csv` , first line). Pipeline 3 (evidence clips) is designed and validated separately and is not built, so no video is kept beyond the 20-second relay window.